Hands-on Activity 6.1: Functions

Functions

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming Logic and Design	**Date Performed: October 7, 2026**
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1. Objectives

This activity aims to develop programming skills using functions.

2. Intended Learning Outcomes (ILOs):

2.1 Create a program in C++ that will add, subtract, divide, multiply for a 2 user-input integer values. Use functions in the program.
2.2 Create a program in C++ that will convert a user-input Farenheight to Celcius and vice versa. Use functions in the program.
2.3 Create a program in C++ that will convert dollars into pesos and vice versa. Use functions in the program.

3. Materials and Equipment

Wi-Fi and Laptop

4. Procedure

2.1 Create a program in C++ that will add, subtract, divide, multiply for a 2 user-input integer values. Use functions in the program.

2.2 Create a program in C++ that will convert a user-input Farenheight to Celcius and vice versa. Use functions in the program.

2.3 Create a program in C++ that will convert dollars into pesos and vice versa. Use functions in the program.

5. Output

1.

CODE:

```
#include <iostream>
using namespace std;

// Function declarations
int add(int a, int b);
int subtract(int a, int b);
int multiply(int a, int b);
double divide(int a, int b);

int main() {
    int num1, num2; // the two integers entered by the user

    // asks the user for the integers
    cout << "Enter first integer: ";
    cin >> num1;
    cout << "Enter second integer: ";
    cin >> num2;

    // Call each function and display< the result
    cout << "\nResults:\n";
    cout << num1 << "+" << num2 << "=" << add(num1, num2) << endl;
    cout << num1 << "-" << num2 << "=" << subtract(num1, num2) << endl;
```

```
        cout << num1 << "*" << num2 << "=" << multiply(num1, num2) << endl;

        // for division dividing by zero is undefined, so we check the devisor first
        if (num2 != 0) {
                cout << num1 << " / " << num2 << " = " << divide(num1, num2) << endl;
    } else {
         cout << num1 << " / " << num2 << " = Error: cannot divide by zero" << endl;
        }

        return 0;
}

// Function definitions
// Adds two integers and returns the sum
int add(int a, int b) {
    return a + b;
}

// Subtracts b from a and returns the difference
int subtract(int a, int b) {
    return a - b;
}

// Multiplies two integers and returns the product
int multiply(int a, int b) {
    return a * b;
}

// Divides a by b and returns the quotient
double divide(int a, int b) {
    // Cast to double so the decimal part is kept
    return static_cast<double>(a) / b;
}
```

OUTPUT:

```
Enter first integer: 49
Enter second integer: 7

Results:
49+7=56
49-7=42
49*7=343
49 / 7 = 7

--------------------------------
Process exited after 11.81 seconds with return value 0
Press any key to continue . . .
```

2.

CODE:

```
#include <iostream>
using namespace std;

// Function prototypes (declarations)
double fahrenheitToCelsius(double f);   // C = (F - 32) * 5/9
double celsiusToFahrenheit(double c);   // F = (C * 9/5) + 32

int main() {
   int choice;     // menu choice entered by the user
   double value;   // temperature entered by the user

   // Display the menu
   cout << "Temperature Converter\n";
   cout << "1. Fahrenheit to Celsius\n";
   cout << "2. Celsius to Fahrenheit\n";
   cout << "Enter your choice (1 or 2): ";
   cin >> choice;

   if (choice == 1) {
      // Fahrenheit -> Celsius
      cout << "Enter temperature in Fahrenheit: ";
      cin >> value;
      cout << value << " F = " << fahrenheitToCelsius(value) << " C" << endl;
   } else if (choice == 2) {
      // Celsius -> Fahrenheit
      cout << "Enter temperature in Celsius: ";
      cin >> value;
      cout << value << " C = " << celsiusToFahrenheit(value) << " F" << endl;
   } else {
      // Anything other than 1 or 2 is not a valid option
      cout << "Invalid choice. Please run the program again." << endl;
   }

   return 0;
}

// Function definitions
// Converts a Fahrenheit temperature to Celsius
double fahrenheitToCelsius(double f) {
   // Use 5.0 and 9.0 (decimals) so the division is not truncated
   return (f - 32.0) * 5.0 / 9.0;
}

// Converts a Celsius temperature to Fahrenheit
double celsiusToFahrenheit(double c) {
   return (c * 9.0 / 5.0) + 32.0;
}
```

OUTPUT:

```
Temperature Converter
1. Fahrenheit to Celsius
2. Celsius to Fahrenheit
Enter your choice (1 or 2): 2
Enter temperature in Celsius: 67
67 C = 152.6 F

--------------------------------
Process exited after 15.85 seconds with return value 0
Press any key to continue . . .
```

3.
CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;

// Exchange rate: pesos per 1 US dollar.
// This is only an approximation; change it to match the current rate.
const double RATE = 62.86;

// Function prototypes (declarations)
double dollarsToPesos(double dollars);   // pesos = dollars * RATE
double pesosToDollars(double pesos);     // dollars = pesos / RATE

int main() {
    int choice;      // menu choice entered by the user
    double amount;   // amount of money entered by the user

    // Show money values with exactly 2 decimal places
    cout << fixed << setprecision(2);

    // Display the menu
    cout << "Currency Converter (1 USD = " << RATE << " PHP)\n";
    cout << "1. Dollars to Pesos\n";
    cout << "2. Pesos to Dollars\n";
    cout << "Enter your choice (1 or 2): ";
    cin >> choice;

    if (choice == 1) {
        // Dollars -> Pesos
        cout << "Enter amount in dollars: ";
        cin >> amount;
        cout << "$" << amount << " = PHP " << dollarsToPesos(amount) << endl;
    } else if (choice == 2) {
        // Pesos -> Dollars
        cout << "Enter amount in pesos: ";
        cin >> amount;
        cout << "PHP " << amount << " = $" << pesosToDollars(amount) << endl;
    } else {
```

```
        // Anything other than 1 or 2 is not a valid option
        cout << "Invalid choice. Please run the program again." << endl;
    }

    return 0;  // program ended successfully
}

// Function definitions

// Converts US dollars to Philippine pesos
double dollarsToPesos(double dollars) {
    return dollars * RATE;
}

// Converts Philippine pesos to US dollars
double pesosToDollars(double pesos) {
    return pesos / RATE;
}
```

OUTPUT:

```
Currency Converter (1 USD = 62.86 PHP)
1. Dollars to Pesos
2. Pesos to Dollars
Enter your choice (1 or 2): 1
Enter amount in dollars: 1
$1.00 = PHP 62.86

--------------------------------
Process exited after 9.183 seconds with return value 0
Press any key to continue . . .
```

6. Conclusion

Through these three programs, I learned how functions break a problem into small, reusable parts, such as separate functions for addition, temperature conversion, and currency conversion. I also saw how details like casting to double for division, checking for division by zero, and using a constant for the exchange rate make a program more accurate and easier to update. Overall, the exercises strengthened my understanding of function prototypes, parameters, and return values, which I can apply to larger C++ programs.

7. Rubric

Rubric for SO 7 (14)

Criteria	Ratings					
SO 7 PI 1 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge
SO 7 PI 2 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently
SO 7 PI 3 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources
SO 7 PI 4 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted